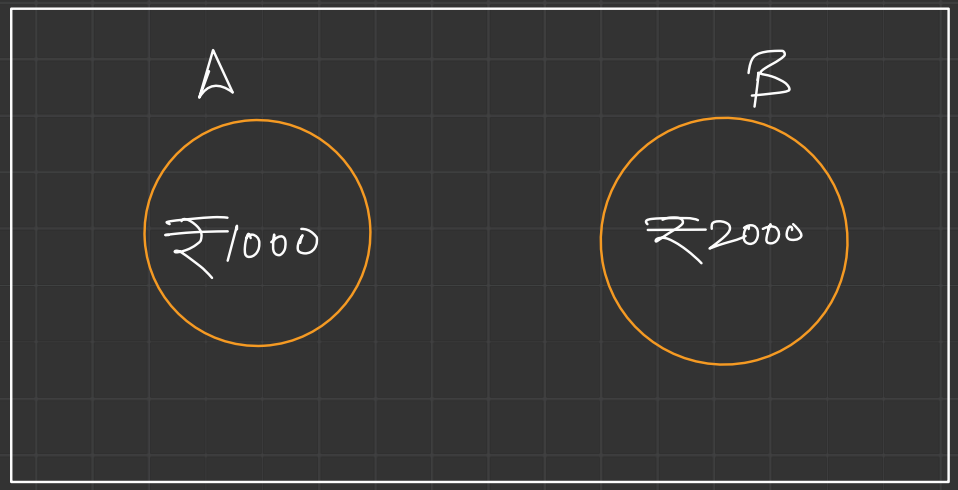
Lec-12

Transaction :-

ABC Bank



System ₹3000

Task 1 → ₹50 from Bank A/C A to A/C B

T_1:

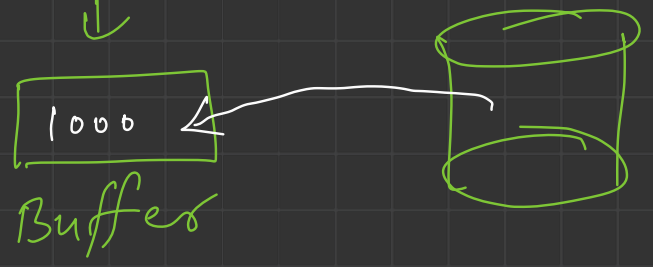
```
read(A)
A := A - 50;
Write(A);
read(B)
B := B + 50;
write(B)
```

6 steps

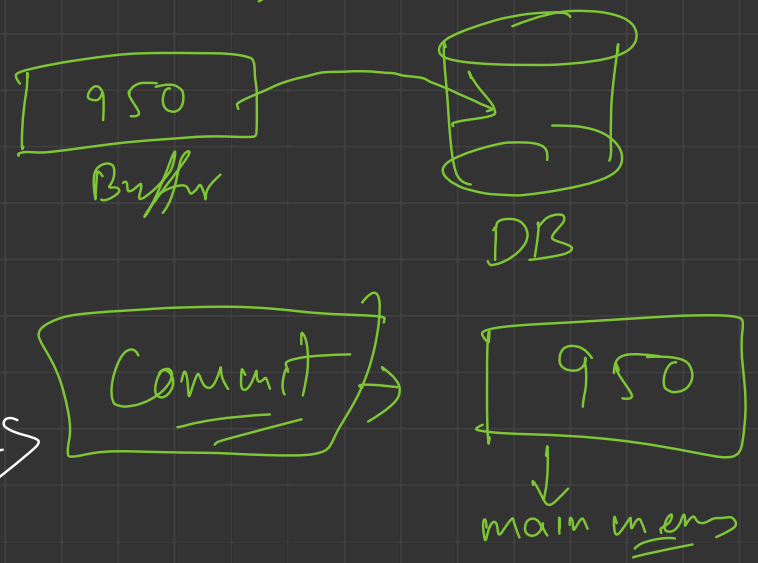
Atomic → they are considered to be a single Task.

ACID Properties

read(A)



write(A);



T_i : read(A)
A := A − 50;
write(A);
read(B);
B := B + 50;
write(B);

① Atomicity

→

A	B
read 1000	read 2000
950 = 1000 − 50	
write(A)	
950	
	read 2000
	2000 + 50
	write(2050)
	2050
950.	

⇒ transaction → successful.

→ fail → old state recovery.

DB → maintain → old state.

↓ intermediate state.

↓ success ✗.

Rollback old states

② Consistency

③ Isolation :-

Banking system

T_1 T_2 T_3 T_4

$t=0 \Rightarrow T_1$ (Google Pay) $t=10 = T_2$ (Netbanking)

read(A) → 1000 ⟶ read₁(A) → 1000.

A := A − 50 = 950.

1000 − 50 = 950.

A = 950

write(950)

B ⟶

B ⟶ 50 + 50 = 100

(4) Durability ⇒

$T_1 \rightarrow$ success

↓

all 6 step → success

↓

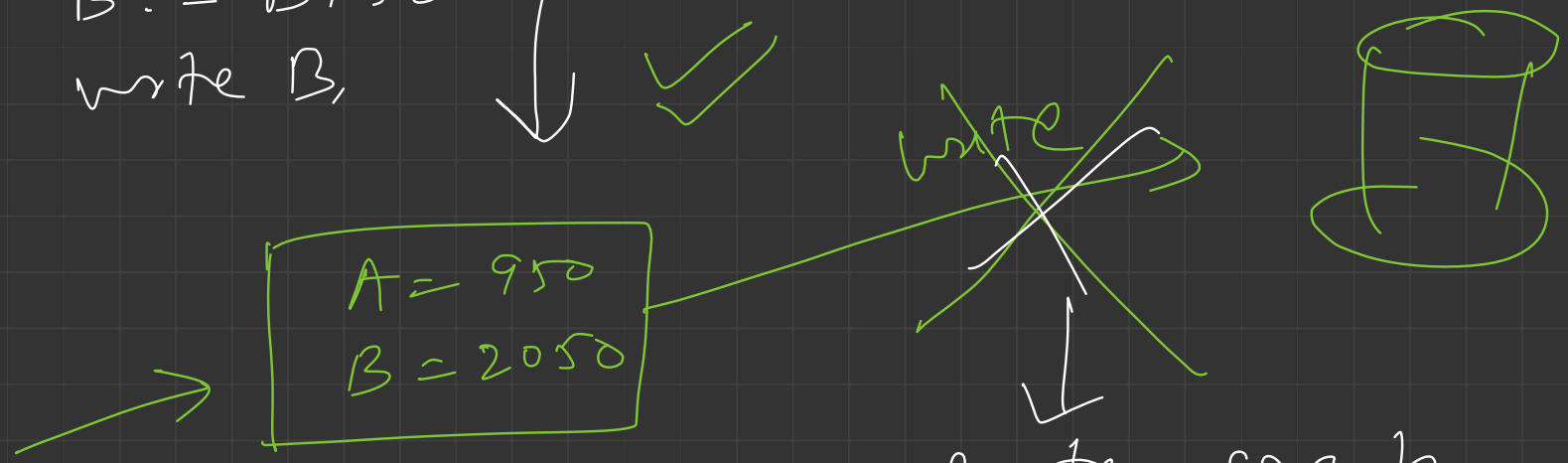
DB updation permanent (persistent)

↓

even if there is a system failure after T completes execute.

→ recovery - mgmt component.

read(A)
A := A - 50 → log (50)
write(A) → log (write A)
Read(B) → (Read)
B := B + 50
write B,



* States of Transaction (life cycle)

How to Implement Atomicity?

Lec - 13